

A 28-nm Optical Flow Estimation Accelerator with Redundancy Speculation, Bit-Width-Aware Compression and Similarity Detection

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Abstract—Optical flow estimation with event-based cameras could capture per-pixel motion field efficiently for intelligent scenarios. However, numerous redundant operations and large unnecessary external-memory-access (EMA) hinder its expected high-efficiency. To alleviate this, we demonstrate an accelerator with redundancy speculation, bit-width-aware compression and similarity detection, decreasing overall operations and EMA to 0.20x and 0.08x, while reducing corresponding energy and latency cost to 0.12x and 0.19x, respectively. In evaluation, an EMA-included energy efficiency of 14.07 TOPS/W is achieved.

Keywords—optical flow estimation, event-based cameras, redundant operations, weight compression, external memory access

I. INTRODUCTION

Optical flow estimation is a classic vision task that captures precise motion field in real time. Through generating an output optical flow field with the same size of input, per-pixel velocity could be obtained and then be utilized in downstream tasks, such as ego-motion estimation, segmentation, simultaneous localization and mapping (SLAM) [1], [2], [3]. On one hand, with the emerging of edge intelligent scenarios that have limited power budget, for example, autonomous driving or flying, embodied intelligence, etc., providing an efficient optical flow estimation becomes more than essential. On the other hand, these edge scenarios require a high-speed and accurate motion capturing, which can hardly be achieved by traditional frame-based cameras due to their fixed/low temporal resolution [4]. Therefore, event-based cameras with low power consumption and high temporal resolution, such as Dynamic Vision Sensors (DVS), have drawn widespread attention in optical flow estimation [5], [6].

To extract spatial-temporal information from spike streams of event-based cameras and finally generate optical flow results, novel neural network models have been proposed [6], [7], [8]. Among these networks, a Hybrid U-Net architecture [7] exhibits outstanding results in Average Endpoint Error (AEE) evaluation, which calculates the distance between prediction results and the ground truth. As illustrated in Fig. 1, this U-Net architecture exploits Spiking Neural Networks (SNN) to process spatial-temporal information, and Artificial Neural Networks (ANN) to estimate real-valued optical flow results. Through multi-level SNN down-sampling and ANN up-sampling, Hybrid U-Net achieves an AEE of <1 pixel on MVSEC dataset [5].

However, when implementing this Hybrid U-Net in hardware, problems from three aspects are hindering its expected high-efficiency, as depicted in Fig. 2. Firstly, non-linear functions (MaxPool and ReLU functions) in the network introduce a large proportion of redundant

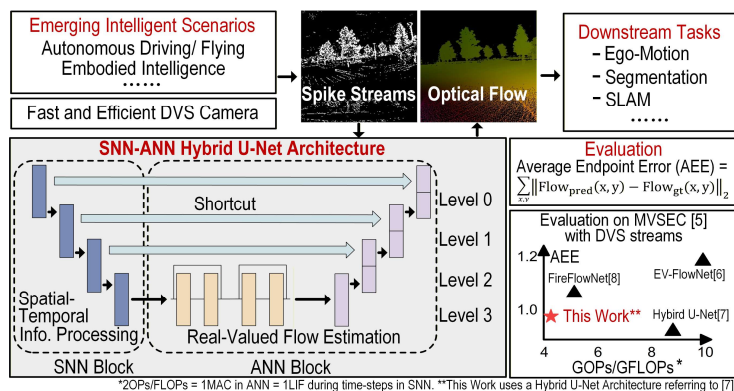


Fig. 1. Optical flow estimation with Hybrid U-Net.

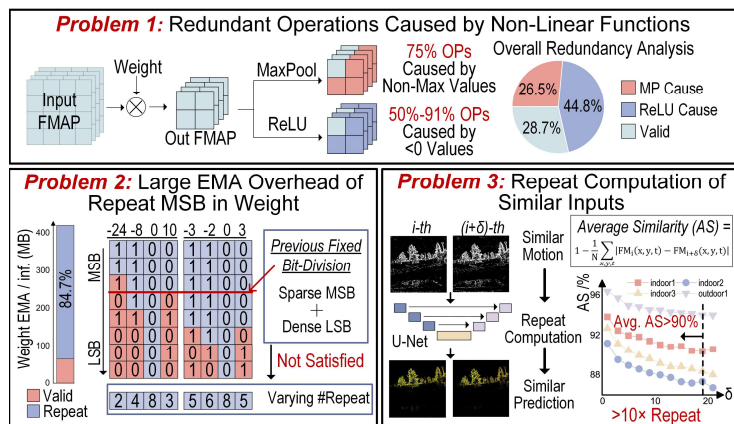


Fig. 2. Problems hindering high-efficiency.

operations. As MaxPool and ReLU abandon all non-maximum and negative values respectively, these abandoned values do not participate in computations of the next layer. Thus, operations calculating out the abandoned values are redundant, taking up 71.3% of all operations. Secondly, large external-memory-access (EMA) overhead of repeat sign bits in weight parameters are witnessed. Repeat bits at MSB end have varying widths, and contribute to 84.7% of overall weight EMA. Previous methods [9] divide weights into two parts of fixed bit-widths and save separately, but can't satisfy varying width requirements. Finally, adjacent input streams may exhibit minimal change when the motion state is steady or the background is empty, leading to similar computations. Analysis on 4 sub-datasets of MVSEC show that average similarity of input streams could reach over 90% with an average input interval $\delta > 10$, indicating non-negligible repeat computations.

This work was supported by the National Key Research and Development Program of China (Grant No. 2024YFB4405501).

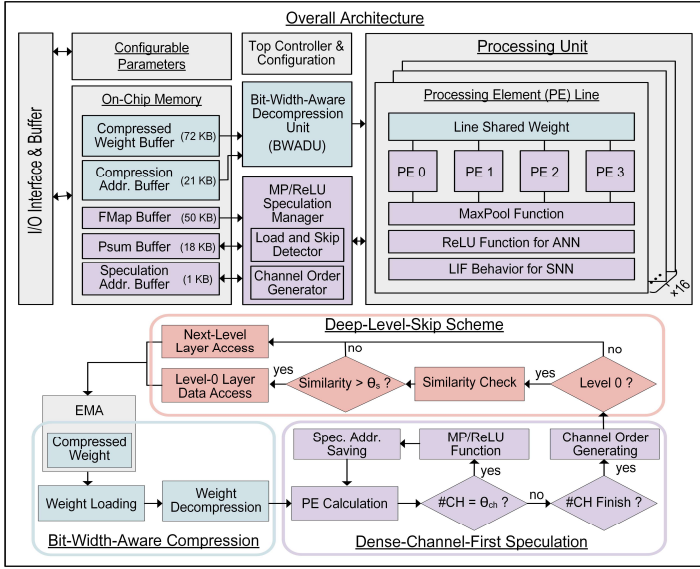


Fig. 3. Overall architecture (top) and system workflow (bottom).

Faced with redundant operations caused by non-linear functions, large EMA overhead of repeat sign bits and similar input streams in optical flow estimation, this paper demonstrates an accelerator with following three features to alleviate these problems. Firstly, Dense-Channel-First Speculation is proposed to speculate redundant locations caused by MaxPool and ReLU. Secondly, Bit-Width-Aware Compression (BWAC) and on-chip decompression are implemented to satisfy varying bit-width demands and reduce EMA cost. Finally, based on dynamic input similarity detection, Deep-Level-Skip Scheme (DLSS) for the multi-level U-Net is proposed to decrease repeat computations for similar input streams. With these features, overall operations and EMA are decreased to 0.20 $\times$ and 0.08 $\times$, respectively. Measurement results show that corresponding energy and latency cost are reduced to 0.12 $\times$ and 0.19 $\times$, achieving an on-chip energy efficiency of 35.80 TOPS/W and an EMA-included energy efficiency of 14.07 TOPS/W in evaluation of MVSEC dataset.

II. PROPOSED ACCELERATOR

A. Overall Architecture and Workflow

Fig. 3 shows the overall architecture of our proposed accelerator. It consists an I/O interface, a top controller and configuration module, an on-chip memory, a bit-width-aware decompression unit (BWADU), a Speculation Manager, and a processing unit. The on-chip memory stores compressed weight, compression address, feature map (FMap), partial sum (Psum), and speculation address in SRAM-implemented buffers. Data access of on-chip memory is controlled by BWADU and the Speculation Manager, and processed data are sent to the processing unit for computation, which includes 16 processing element (PE) lines in total. Each PE line contains a line-shared weight decompressed from BWADU, modules for MaxPool/ReLU/LIF functions, 4 PEs that could perform both multiply-and-accumulation (MAC) operations in ANN and accumulation (AC) operations in each time-step of SNN. The system workflow is also illustrated in Fig. 3:

First, compressed weight parameters and corresponding addresses are loaded through EMA, and decompressed online.

Then, redundancy speculation for MaxPool/ReLU is carried out during PE calculations. As input feature map is fed into PE lines in the

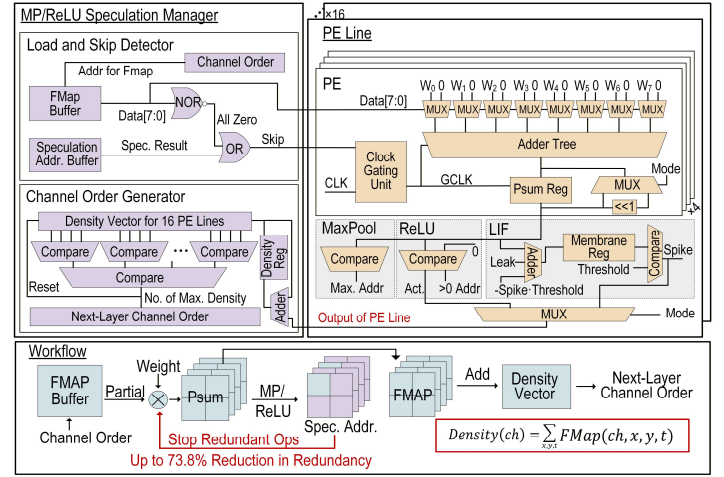


Fig. 4. Implementation to support speculation, and its workflow.

order of Channel Density, input channels used in calculation are continuously increasing. Once the number of input channels (#CH) is over the threshold θ_{ch} , MaxPool/ReLU function is performed immediately to speculate non-maximum or negative addresses. For the rest channels, computations contributing to these speculated addresses are stopped. Later, a new channel order is generated by sorting Channel Density, and will be used for computations in the following layer.

Finally, whether to skip deep levels of U-Net is determined based on similarity check. If current layer is Level 0 and similarity is over the threshold θ_s , only Level 0 will be computed, which can be considered as a shallower U-Net. Otherwise, the whole U-Net will be computed.

B. Dense-Channel-First Speculation

As detailed in Fig. 4, Dense-Channel-First Speculation is mainly implemented by PE lines and the Speculation Manager. Each PE line resembles an output channel and compute independently. And every PE could handle 8 inputs simultaneously, realizing 8 single-bit input $\times$ 8-bit weight operations through MUXs and an adder tree. PE's results can be interpreted as AC operations in a time-step of SNN. Also, they can be considered as part of MAC operations in ANN. Through recurrently shifting Psum and adding, full 8-bit $\times$ 8-bit operations are realized. MaxPool, ReLU and LIF functions are then implemented with simple comparators and adders as shown in the figure.

As for the Speculation Manager, it consists a Load and Skip Detector to stop computing, and a Channel Order Generator to sort Channel Density. Load and Skip Detector detects whether input data are all zero, or whether current computing addresses are speculated to be redundant. If any situation is detected, a Skip signal would trigger clock gating and stop corresponding computations in PEs. Channel Order Generator firstly accumulates output data from 16 PE Lines (output channels), and takes the accumulation results as Channel Density. Then, the sorting of Channel Density is realized through stacked comparators in several rounds. Each round the stacked comparators select out current maximum density value, save its channel number, and reset it to zero. Then, another round of selecting begins until requiring channel order is obtained. With speculation, redundancy caused by MaxPool and ReLU are reduced by up to 73.8% on MVSEC.

C. Bit-Width-Aware Compression and Decompression

Fig. 5 depicts the feature of Bit-Width-Aware Compression and its on-chip decompression's implementation. In compression, adjacent

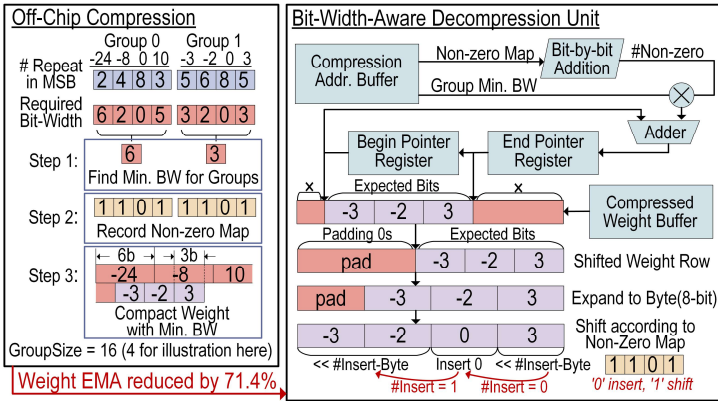


Fig. 5. BWAC and its decompression module BWADU.

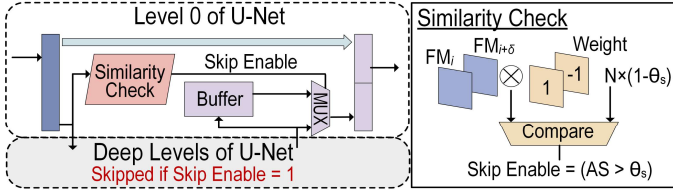


Fig. 6. DLSS and its implementation.

weight parameters are grouped together, and group-wise required minimum bit-widths (Min. BW) can be determined, which allows a varying bit-width. For example, Group 0 requires 6 bits to cover a range from -24 to 10, while Group 1 only demands a 3-bit width to save weight values from -3 to 3. Moreover, to further reduce weight overhead, zero values are expected to be squeezed. Therefore, a bit-map (Non-zero Map) is used to record locations of non-zero values. Finally, weights are arranged in a compact format, i.e. aligning in a row one by one with their Min. BW. If exceeding the bit-width of on-chip memory, then a new row will be used for arrangement.

Compressed weights and their address information (Min. BW and Non-zero Map) are saved separately, and decompressed by BWADU module. At first, Min. BW and Non-zero Map are used to calculate the total bit-width of a group. Then, Begin Pointer Register and End Pointer Register are updated by the total bit-width. These two registers point out where current group's parameters begin/end in weight rows. And according to this, expected bits are picked out, while other bits are truncated, as shown in Fig. 5. Later, expected bits are expanded to original widths (1-Byte) by bit-shifting. Finally, weight parameters are recovered to their original order in a group according to Non-zero Map. From right to left of Non-zero Map, if "0" is encountered, a zero value needs to be inserted, and #insert increases; otherwise if "1" is detected, expanded bits are left-shifted by #insert bytes. With this Bit-Width-Aware-Compression and Decompression, overall weight EMA cost could be reduced by 71.4%.

D. Deep-Level-Skip Scheme

According to the model in Fig. 1, Level 0 of U-Net includes an SNN layer and an ANN layer, thereby can be considered as a shallower U-Net. Fig. 6 illustrates our proposed Deep-Level-Skip Scheme. Level 0 should be computed for all input streams, while deep levels of U-Net are triggered by similarity check, which calculates Average Similarity (AS) between two feature maps:

$$AS = 1 - \frac{1}{N} \sum_{x,y,t} |FM_i(x,y,t) - FM_{i+\delta}(x,y,t)|, \quad (1)$$

where FM_i refers to i -th feature map, (x,y,t) refers to indices of feature maps, and N refers to the total number of values. Equation (1)

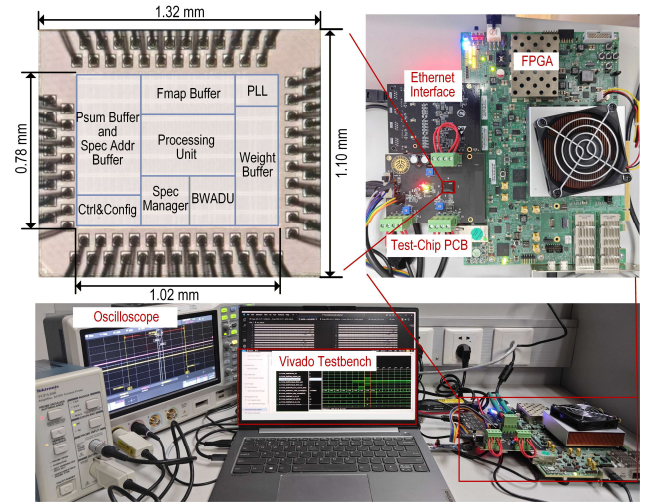


Fig. 7. Die photograph and measurement setup.

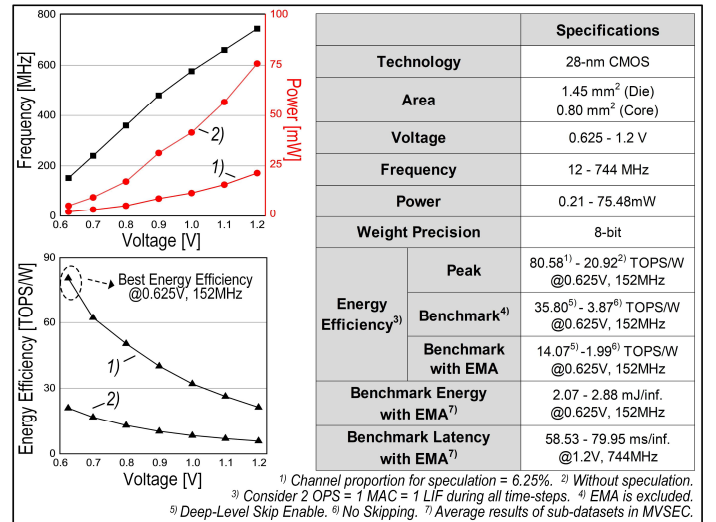


Fig. 8. Voltage scaling and specification summary.

is primarily implemented by configuring on-chip memory and PE lines: FM_i and $FM_{i+\delta}$ are concatenated as input, and weight parameters are set to be 1/-1. Then through comparing with a threshold, Skip Enable signal is generated.

III. MEASUREMENT RESULTS

A. Setup and Specifications

We implemented the proposed accelerator in a 28-nm CMOS technology. Fig. 7 shows the die photograph and the measurement setup. The total die area is 1.45 mm², with the core area of 0.80 mm². An FPGA is used as an interface between the host computer and test-chip PCB. Fig. 8 shows the specifications of our test-chip. It could operate under the frequency of 12-744 MHz at the supply voltage of 0.625-1.2 V, with a total power consumption of 0.21-75.48 mW. The best energy efficiency is achieved at 0.625V, 152 MHz, reaching 80.58-20.92 TOPS/W varying with the channel proportion used for speculation. When testing on 4 sub-datasets of MVSEC benchmark [5], on-chip energy efficiency could achieve 35.80 TOPS/W with DLSS enabled, and the overall energy efficiency considering EMA could reach 14.07 TOPS/W. Here, EMA overhead is estimated with 3.7 pJ/bit

TABLE I. METRICS ON SUB-DATASETS OF MVSEC

	indoor1	indoor2	indoor3	outdoor1 ¹⁾	Mean
Baseline AEE (INT8)	0.84	1.32	1.14	0.52	0.96
AEE w/ Features (INT8)	0.87	1.35	1.17	0.56	0.99
AEE Degradation	0.03	0.03	0.03	0.04	0.03
Channel Proportion for Speculation	33.41%	33.41%	59.68%	26.23%	38.18%
MP/ReLU Redundancy	0.53×	0.57×	0.82×	0.26×	0.54×
DLSS Interval (5)	15.77	9.38	6.71	11.96	10.95
Overall Operations	0.18×	0.20×	0.25×	0.17×	0.20×
EMA	0.08×	0.09×	0.10×	0.08×	0.08×
Energy Cost ²⁾	0.10×	0.12×	0.14×	0.11×	0.12×
Latency Cost ²⁾	0.17×	0.21×	0.24×	0.19×	0.19×

¹⁾ 800 input samples are considered, referring to [6],[7],[8]. ²⁾ EMA overhead is included.

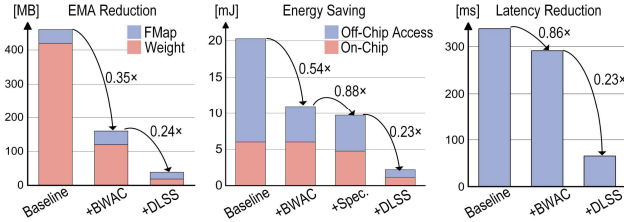


Fig. 9. Analysis on EMA, energy and latency reduction.

and 6.4 GB/s, both based on LPDDR3 SDRAM [10]. Overall energy and latency cost per inference on 4 sub-datasets of MVSEC are 2.07-2.88 mJ and 58.53-79.95 ms, respectively.

B. Benchmark Evaluation

To further analyze the impact of three features in our accelerator, several important metrics are listed in Tab. I. Average Endpoint Error (AEE) reflects the accuracy of optical flow estimation. Baseline mean AEE on 4 sub-datasets of MVSEC is 0.96 (quantized to 8-bit weights). With the methods of redundancy speculation, BWAC and DLSS, the mean AEE increases by only 0.03, showing a negligible degradation. Correspondingly, redundancy caused by MaxPool/ReLU functions drops to 0.26× - 0.82×, and overall operations are reduced to 0.17× - 0.25×. Moreover, detailed analysis on contributions of our proposed features is illustrated in Fig. 9. Mean EMA, energy cost and latency cost are all well-optimized, reducing to 0.08×, 0.12× and 0.19× compared to results without these features, respectively.

C. Comparison and Discussion

Tab. II compares our proposed accelerator with prior state-of-the-art ASIC implementations. The proposed Dense-Channel-First Speculation and Deep-Level-Skip Scheme in this work are co-designed with model evaluation, as discussed in Section III.B. Meanwhile, Bit-Width-Aware Compression is proposed to further reduce EMA cost. Prior works in the table [11], [13], [14], [15] have also implemented methods with model co-designing or weight compression, but they are more focused on one aspect. Measurement results show that the peak energy efficiency of our accelerator is 1.56× - 7.77× higher compared to prior works. In benchmark evaluation, models for optical flow estimation are deployed in [15] and our work. Other works have also used similar convolution-based models and targeted at vision-input tasks. In comparison, our accelerator could achieve 1.46× - 2.31× better on-chip energy efficiency, and 1.58× - 3.47× improvement for system energy efficiency with EMA overhead.

IV. CONCLUSION

In this paper, we introduced an accelerator to alleviate problems of

TABLE II. COMPARISONS WITH PRIOR ASIC IMPLEMENTATIONS

	ISSCC'23 [11]	JSSC'25 [12]	ISSCC'25 [13]	CICC'25 [14]	JSSC'21 [15]	This Work
Technology	28nm	28nm	28nm	28nm	40nm	28nm
Paradigms	SNN, ANN	SNN, ANN	ANN	ANN	ANN	SNN, ANN
Co-designed with Model Pruning/Skiping	yes	no	yes	no	yes	yes
Weight Compression	no	no	no	yes	no	yes
Area [mm ²]	20.25	537.98	20.25	0.67	9.00	1.45
Supply Voltage [V]	0.7-1.1	0.625-1.1	0.68-1.0	0.35-0.9	0.6-0.9	0.625-1.2
Frequency [MHz]	50-200	24-600	50-250	6.55-262	180-400	12-744
Power [mW]	32.3-420.4	10-9970	-	0.43-54	40.8-182.7	0.21-75.48
Precision	INT4/8	INT1/2/4/8	INT4/8/12/16	INT8	INT12	INT8
Peak Energy Efficiency ¹⁾ [TOPS/W]	-	10.37-1.34	39.9-8.6	51.8-9.49	-	80.56 ²⁾ -20.92 ³⁾
Benchmark ⁴⁾	ImageNet @ResNet50	ImageNet @VGG-19	MS-COCO @SDXL-Turbo	Customized ⁵⁾	Flying Chair @FlowNet	MVSEC @Hybrid U-Net
Benchmark Energy Efficiency ¹⁾ [TOPS/W]	24.5	-	15.5-14.2	-	-	35.80 ²⁾ -3.87 ³⁾
System Efficiency with EMA ⁶⁾ [TOPS/W]	-	-	-	8.90-2.54	4.05 ⁶⁾	14.07 ⁶⁾ -1.99 ⁷⁾

¹⁾ 20Ps - 1MAC - 1LIF during time-steps. Measured at 0.625V, 152MHz. EMA is excluded. ²⁾ Channel proportion for spec. is 6.25%. BWAC and DLSS enabled. ³⁾ No speculation. ⁴⁾ All are vision-input tasks and convolution-based models. ⁵⁾ Self-generated data. ⁶⁾ BWAC and DLSS enabled. Average channel proportion for spec. is 38.18%. ⁷⁾ DLSS disabled. ⁸⁾ EMA overhead is estimated. ⁹⁾ Average results of layers reported in [15].

redundant operations, large EMA overhead and similar input streams in optical flow estimation. With our proposed Dense-Channel-First Speculation, Bit-Width-Aware Compression and Deep-Level-Skip Scheme, overall operations and EMA are successfully decreased to 0.20× and 0.08×, respectively. Fabricated in 28-nm CMOS technology, measurement results show that corresponding energy and latency cost are reduced to 0.12× and 0.19×, reaching a best system EMA-included energy efficiency of 14.07 TOPS/W on MVSEC dataset.

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